

Databases and SQL

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1 What to Submit

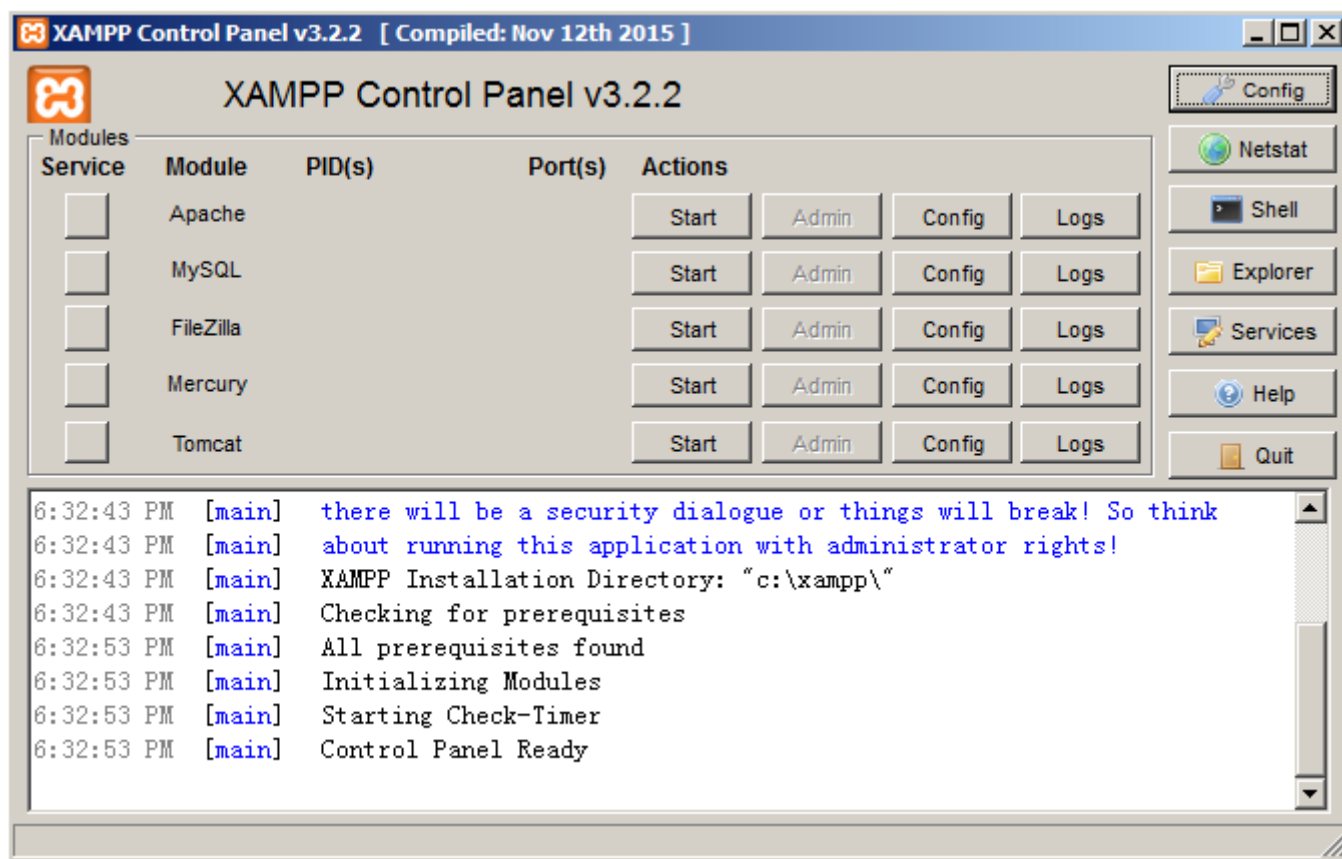
This document contains 10 exercises that you need to do while practicing databases and SQL programming. To submit your answers for these 10 exercises: use the Start menu of Microsoft Windows to start the Notepad software, then write your 10 answers in the Notepad document (give a number from 1 to 10 to each answer), save the document on the desktop of your computer, then upload the document on iSpace at the end of the practice lab.

2 XAMPP

To practice databases and the SQL programming language, you need to have some database management system (DBMS) installed on your computer, plus some graphical interface that allows you to write simple programs in SQL. The XAMPP software package contains everything you need for this lab: the MySQL DBMS and the Apache web server that you can use to send small SQL programs to MySQL using a standard web browser. XAMPP is free software and works on computers running the Microsoft Windows, Apple Mac OS X, or Linux operating systems.

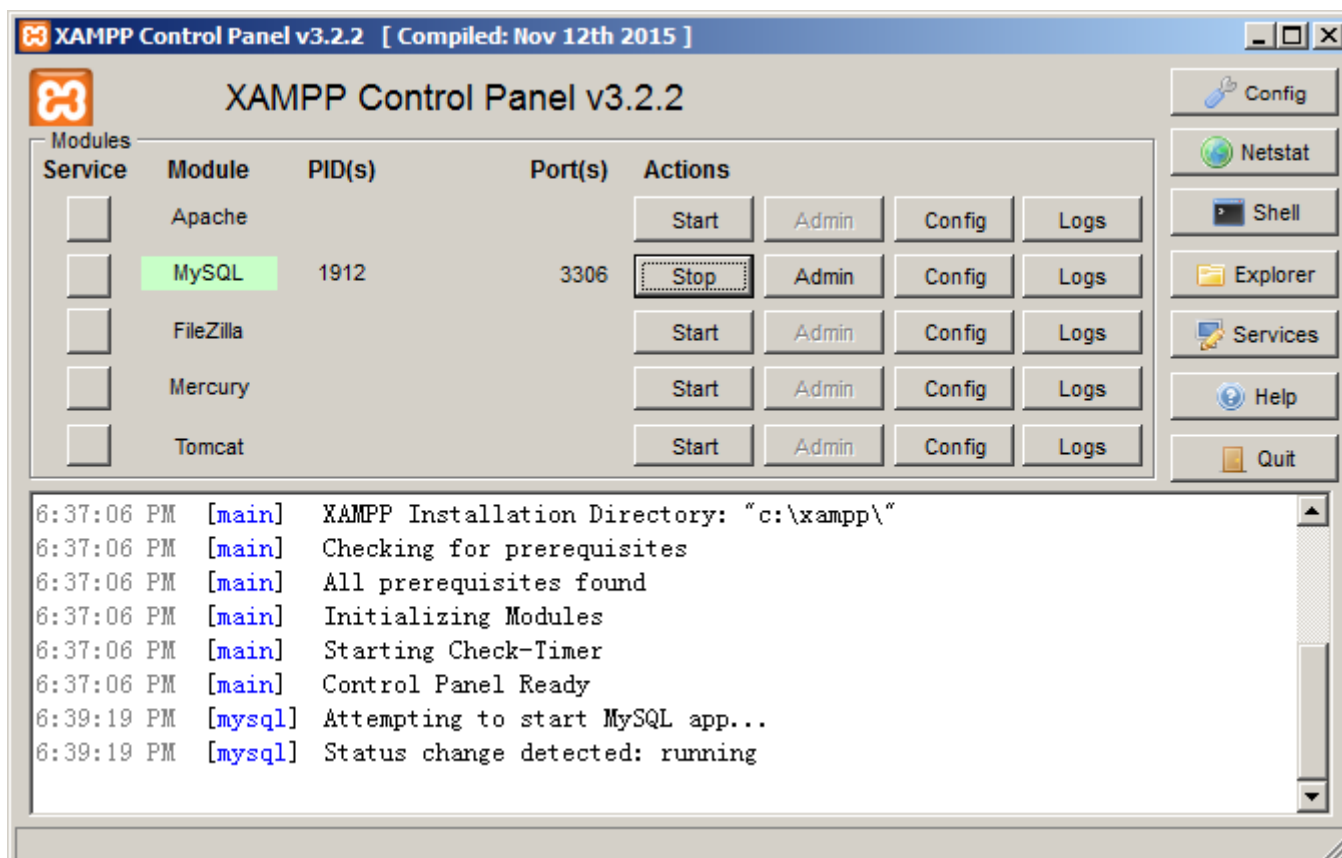
XAMPP is already installed on the UIC computers. If you want to practice on your computer at home then you can download XAMPP for free from <https://www.apachefriends.org/> (download size is a little bit over 100 MB).

Start the XAMPP Control Panel using the Start menu of Microsoft Windows:



2.1 MySQL

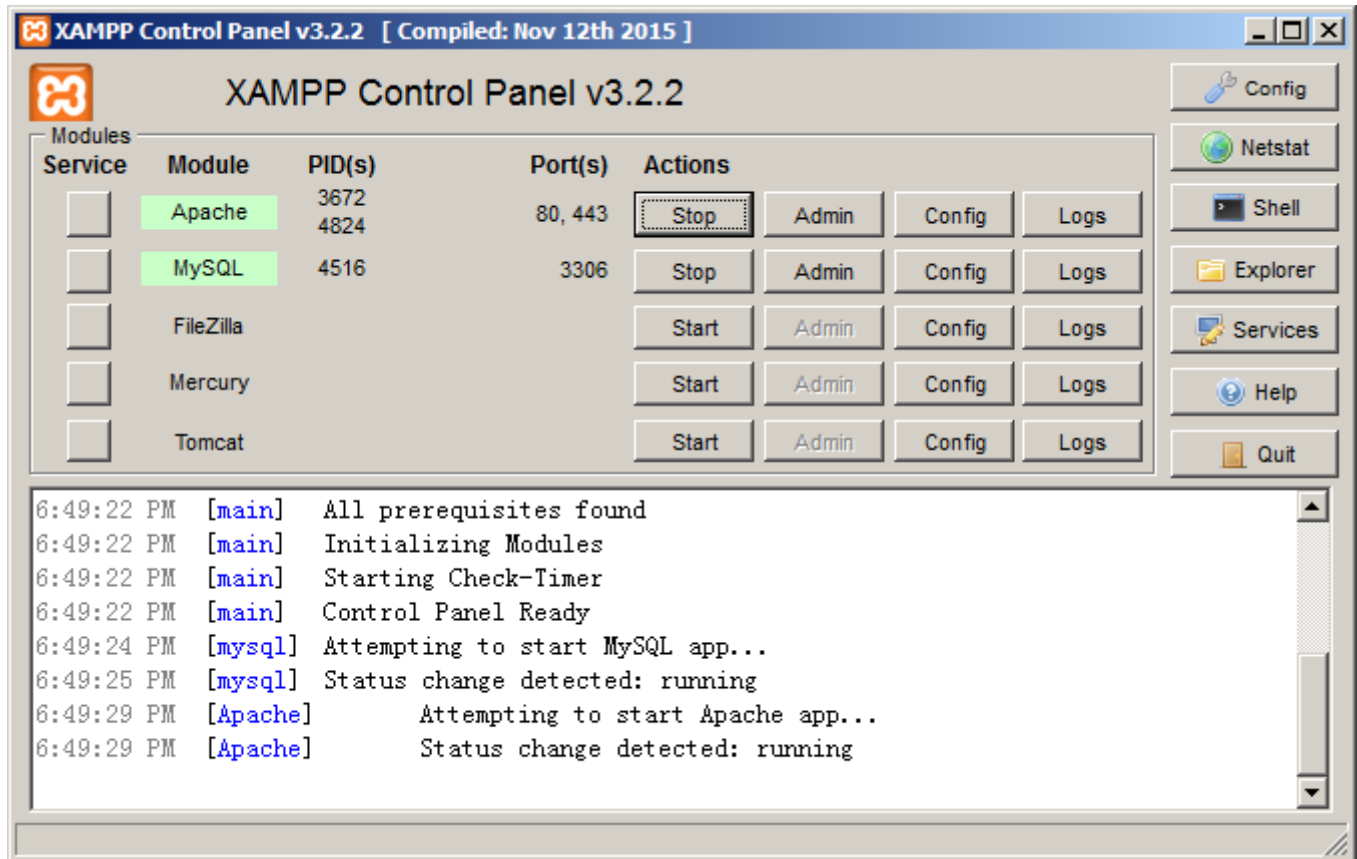
MySQL is a database management system, which allows you to manage and process data in databases. MySQL is included as part of XAMPP and the only thing you need to do is to go to the XAMPP Control Panel and click on the Start button for MySQL. After a few seconds the MySQL name turns green, indicating that MySQL is running correctly:



Later at the end of the practice you need to remember to stop MySQL.

2.2 Apache

Apache is a web server, which you need to run on your computer to help you communicate with the MySQL DBMS. Apache is included as part of XAMPP and the only thing you need to do is to go to the XAMPP Control Panel and click on the Start button for Apache. After a few seconds the Apache name turns green, indicating that Apache is running correctly:



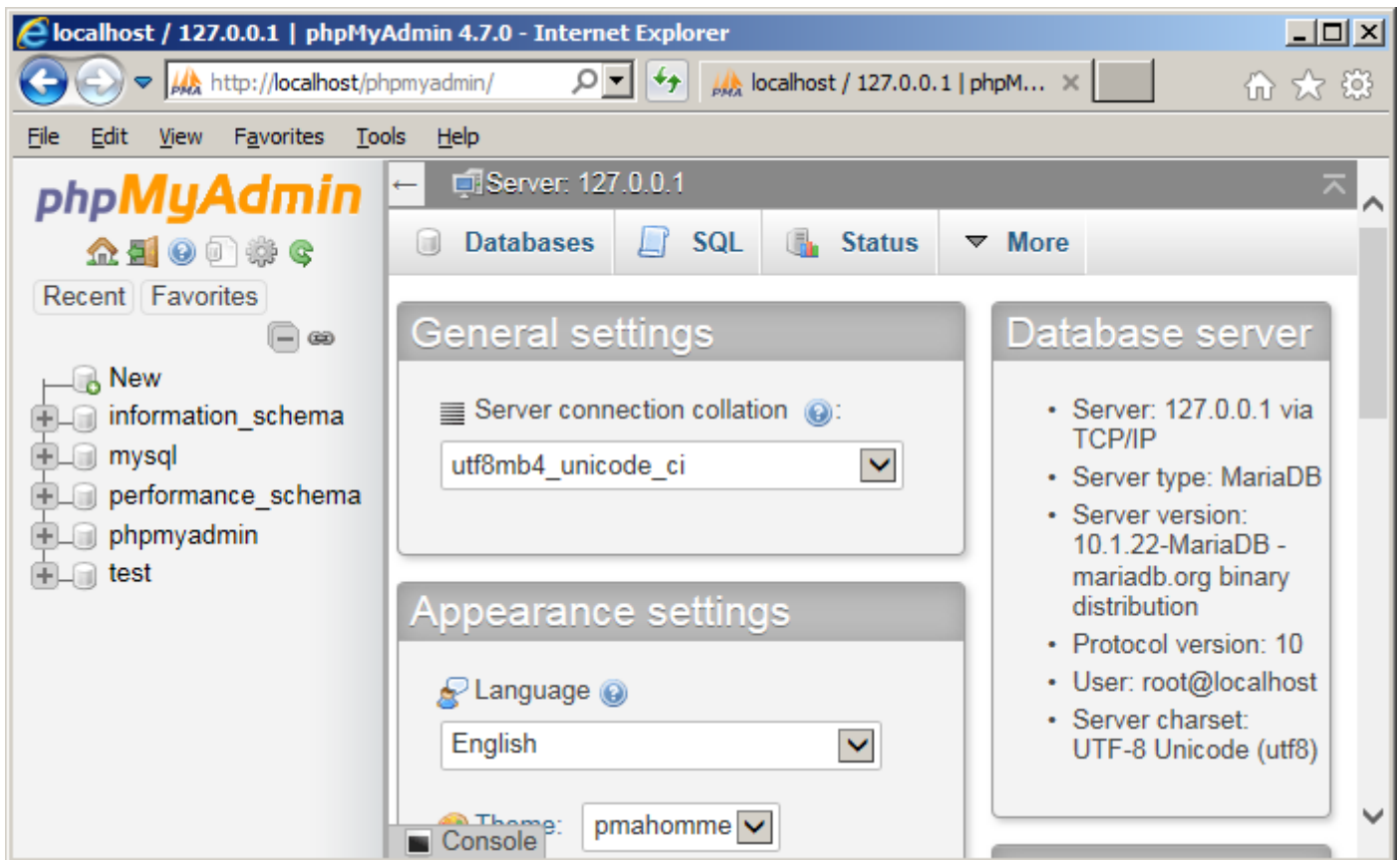
Later at the end of the practice you need to remember to stop Apache.

2.3 Web Browser and phpMyAdmin

It is not possible for you to directly communicate with the MySQL DBMS. Instead you have to communicate with MySQL through the Apache web server, using a web browser. You can use any web browser you want: Internet Explorer, Google Chrome, Mozilla Firefox, etc.

Start your favorite web browser and go to the following web page: <http://localhost/> The name “localhost” always means “this computer” (just like “I” in English always means yourself) so by using the <http://localhost/> web link you then make your web browser connect to the Apache web server running on your computer. You then get in your web browser the “Welcome to XAMPP” web page, which comes from the Apache web server. If you do not get the “Welcome to XAMPP” web page then it means that either you typed the wrong web link or you forgot to start the Apache web server (see above).

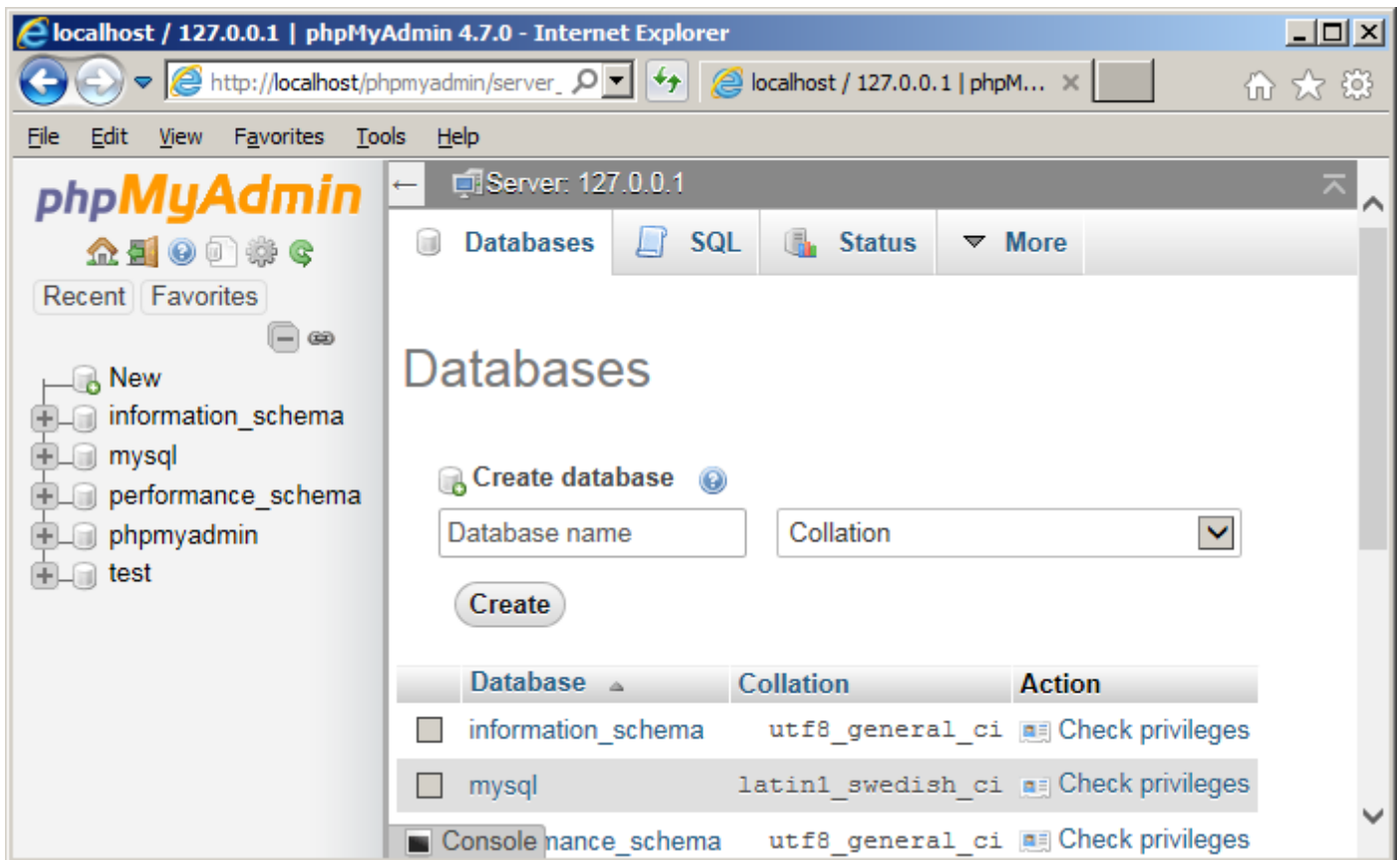
In the top right corner of the “Welcome to XAMPP” web page, click on the phpMyAdmin link. You get the phpMyAdmin web page, which lets you communicate from your web browser through the Apache web server to the MySQL DBMS, all running on the same computer:



Another way to access this web page is to simply click on the Admin button for MySQL in the XAMPP Control Panel: XAMPP will then automatically start Microsoft Windows' default web browser for you and tell the web browser to directly access the web page for phpMyAdmin.

In the left column of phpMyAdmin, you see a list of existing databases which are currently managed by MySQL. If you click on the "+" symbol next to each database, you see a list of the tables defined inside the corresponding database. For this practice, we use the "test" database, which currently does not contain any table.

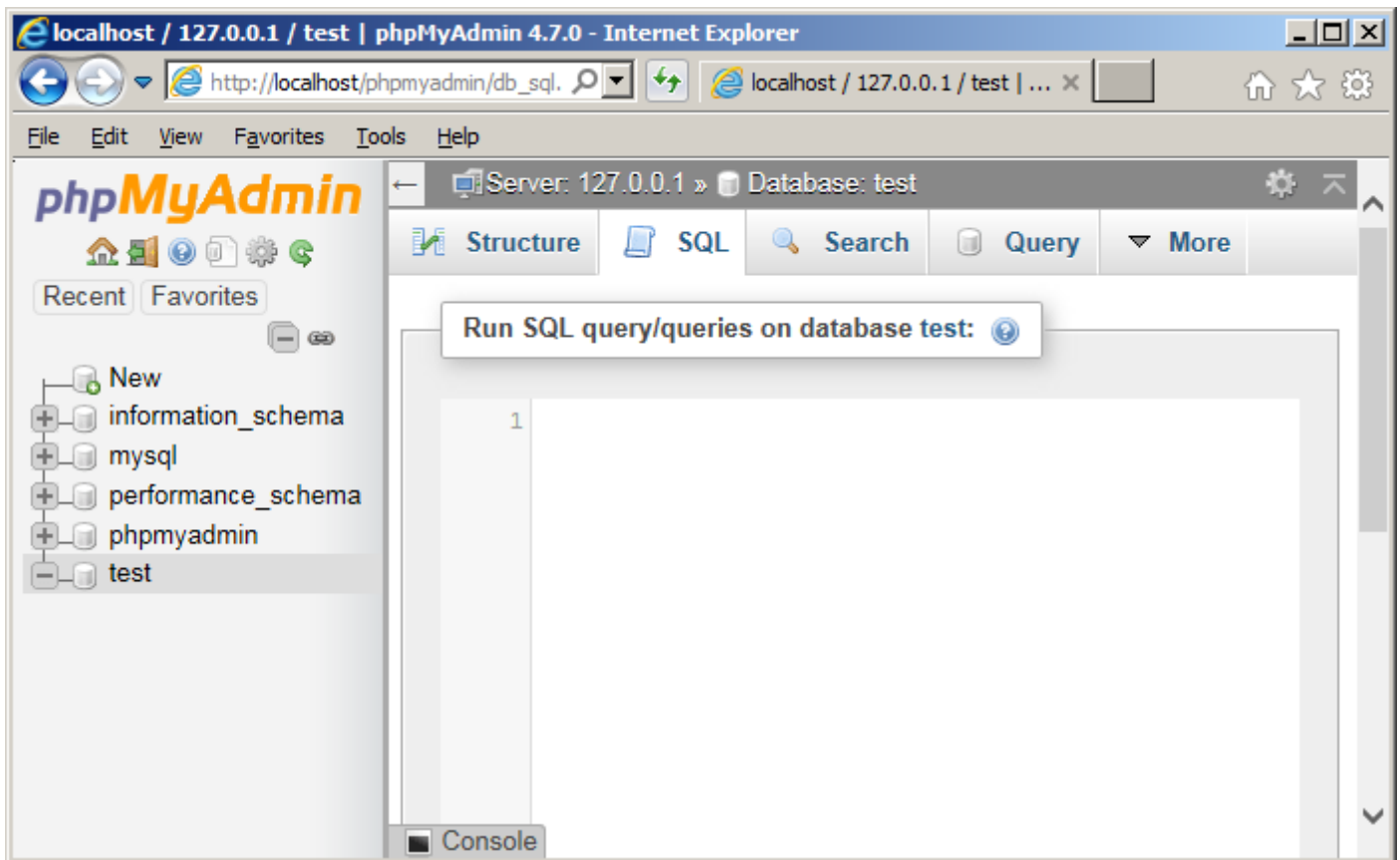
Above the middle column of phpMyAdmin, you see a blue "Databases" link that you can use to manage databases: create new databases, drop (delete) existing databases, etc:



Go down in the list of existing databases and click on the name of the “test” database to select it. Another way to select the “test” database is by clicking on the name of the “test” database in the left column of phpMyAdmin. When you select the “test” database like this, the “Server: 127.0.0.1” link at the top of the web page then becomes “Server: 127.0.0.1 >> Database: test”, which indicates that the “test” database is now selected. At the same time, the “Databases” web page is replaced by a “Structure” web page that shows you all the tables inside the “test” database (which still currently does not contain any table).

To go back to the main phpMyAdmin web page, you can either click on the “Server: 127.0.0.1” link at the top of the web page (not the “Database: test” link), or click on the little home-shaped icon just below the phpMyAdmin logo in the left column of the web page.

Above the middle column of phpMyAdmin, you also see a blue “SQL” link that you can use to write and execute little SQL programs, to create, change, delete, or search data in the “test” database which is currently selected:



Note that the message says “Run SQL query/queries on database test” and that “Server: 127.0.0.1 >> Database: test” appears at the top of the page. If the message you see says “Run SQL query/queries on server 127.0.0.1” and that only “Server: 127.0.0.1” appears at the top of the page then it means that you forgot to select the “test” database. If you have this problem then click on the name of the “test” database in the left column of phpMyAdmin to select it, then click on the blue “SQL” link again, to get the message “Run SQL query/queries on database test”.

3 SQL

3.1 Creating Tables

To practice SQL programming, we want to create two new tables in the “test” database: one table containing student information and one table containing grade information.

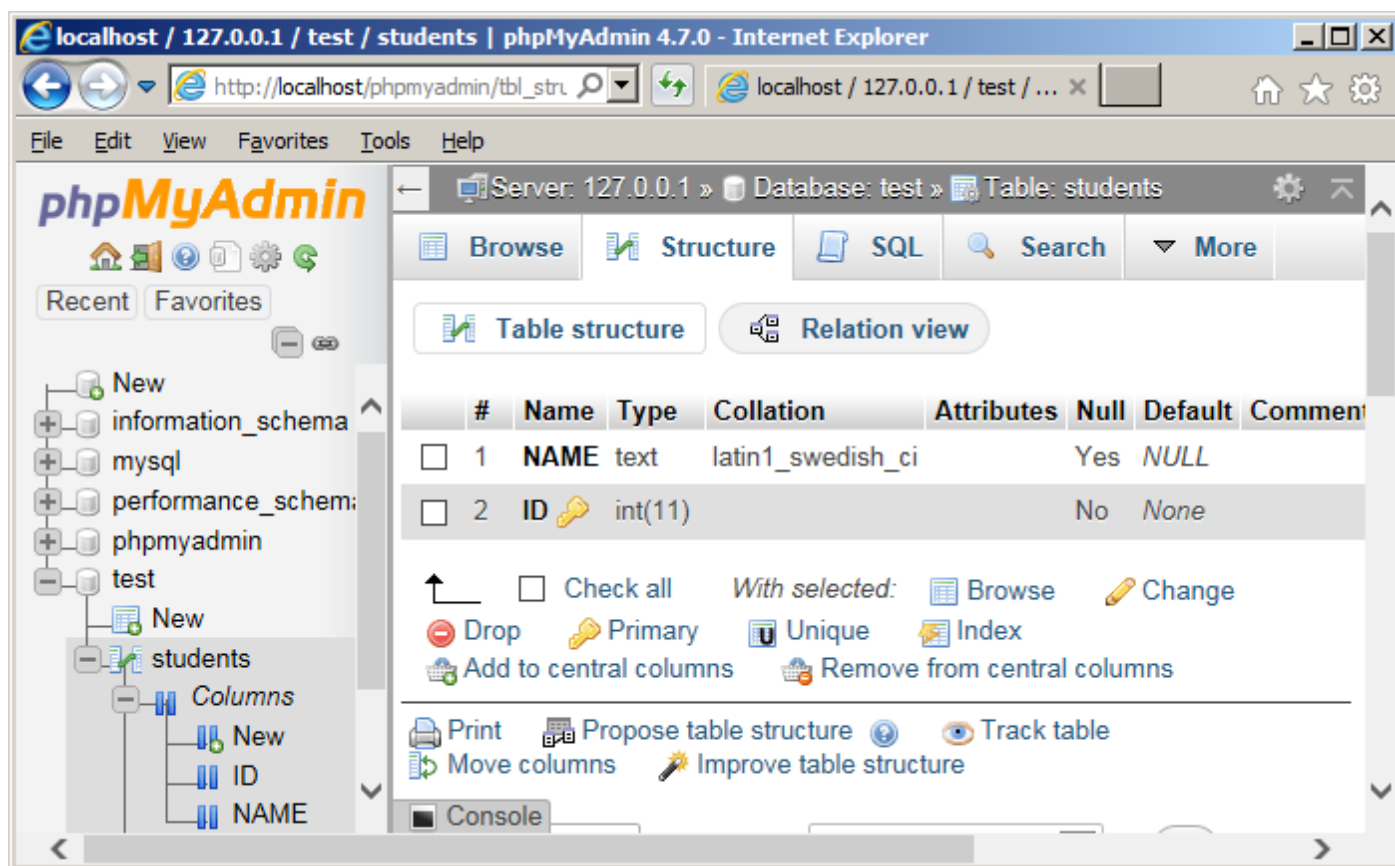
To create a new table using SQL, type the following SQL code into the SQL query box (where the “1” is in the picture above):

```
CREATE TABLE STUDENTS (NAME TEXT, ID INTEGER, PRIMARY KEY(ID));
```

and click on the Go button at the bottom of the web page. This button sends your SQL code to the MySQL DBMS which then executes it. This creates in the “test” database currently selected a new empty table called “students” which has two columns: one column for names, where a name is text, and one column for student ID numbers, where an ID number is an integer. The primary key for the table, which will be used to uniquely identify each student in the table, is defined to be the student ID number.

After executing the code, click on the blue “Structure” link at the top of the page to check the content of the “test” database currently selected. The new web page shows that the “test” database now has a new table called “students”. Click on the small blue “structures” link next to the “students” name of the table to

see the structure of the table. The “Server: 127.0.0.1 >> Database: test” links at the top of the web page then become “Server: 127.0.0.1 >> Database: test >> Table: students”, and below that you can see that the “students” table contains one column named “NAME” of type “text” and one column named “ID” of type “int(11)” (which is the same as “INTEGER”). You can also see some of this information by clicking on the various “+” symbols in the left column of phpMyAdmin:



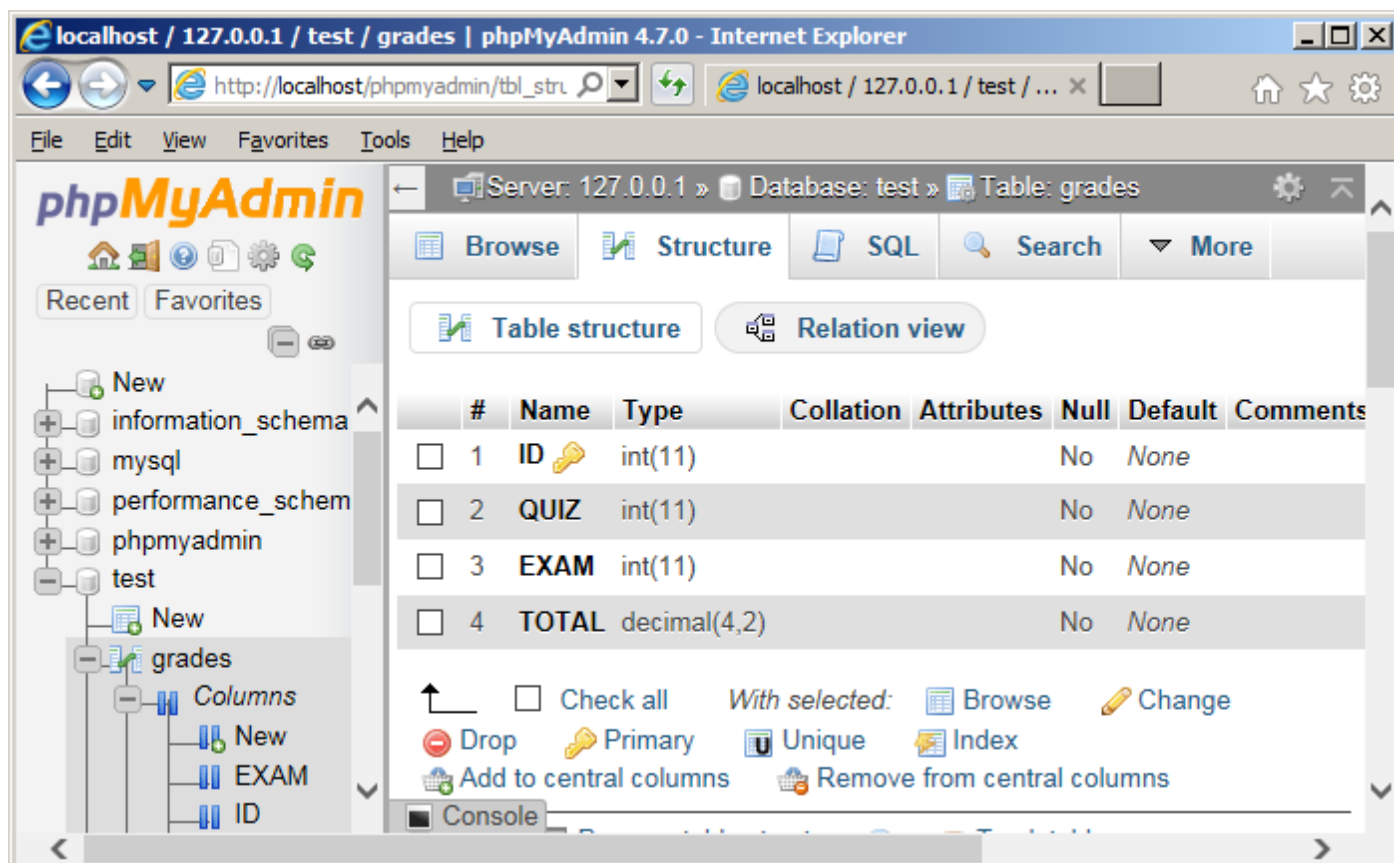
Another way to create a table is to use the phpMyAdmin web interface instead of writing SQL code: click on the “Database: test” link at the top of the web page to go back to the database level, then click on the blue “Structure” link to go back to the list of tables currently defined in the “test” database. At the bottom of this web page you can find a “Create table” box. Type “grades” for the name of the new table, 4 for the number of columns, then click on the Go button.

In the new web page, you then have to specify a name for each of the four columns that you want to have in the new table: ID (for the student ID number), QUIZ (the grade for the quiz), EXAM (the grade for the final exam), and TOTAL (for the student’s total grade at the end of the semester). You also have to specify the type of the data for each of the four columns using the “Type” drop-down menus: in our case the default “INT” type is fine for ID, QUIZ, and EXAM. Use the “DECIMAL” type for the TOTAL, with a “Length/Values” of “4,2” (a total of four digits with two digits after the decimal dot). We also want to use the student ID number as primary key for the new grades table, so, in the web page row for the first table column ID, click on the “Null Index” drop-down menu, select “PRIMARY”, and click Go on the popup window, after which the ID table column will be marked as PRIMARY.

Once you have specified all the necessary information you need for the new “grades” table, click on the “Preview SQL” button at the bottom of the web page. You can then see the SQL code that phpMyAdmin is going to use to create the new table. This code is very similar to the SQL code you typed yourself above when creating the “students” table: the web interface of phpMyAdmin is just a simple way to write SQL code step by step using the keyboard and mouse but inside phpMyAdmin all database actions are still done

using SQL code only. In general when using phpMyAdmin, you always have to choice between writing the SQL code yourself or using various phpMyAdmin web pages to write the code for you. In the rest of this lab we will write the SQL code ourselves, since it is more interesting and fun than using a web interface.

If you are happy with the SQL code for creating the new “grades” table then click on the “Save” button at the bottom of the web page. This sends the SQL code created by phpMyAdmin to the MySQL DBMS which executes it and creates the table. You can see the structure of the new table in the new web page you get in your web browser, and some of the same information is also shown in the left column of phpMyAdmin when you click on the various “+” symbols there:



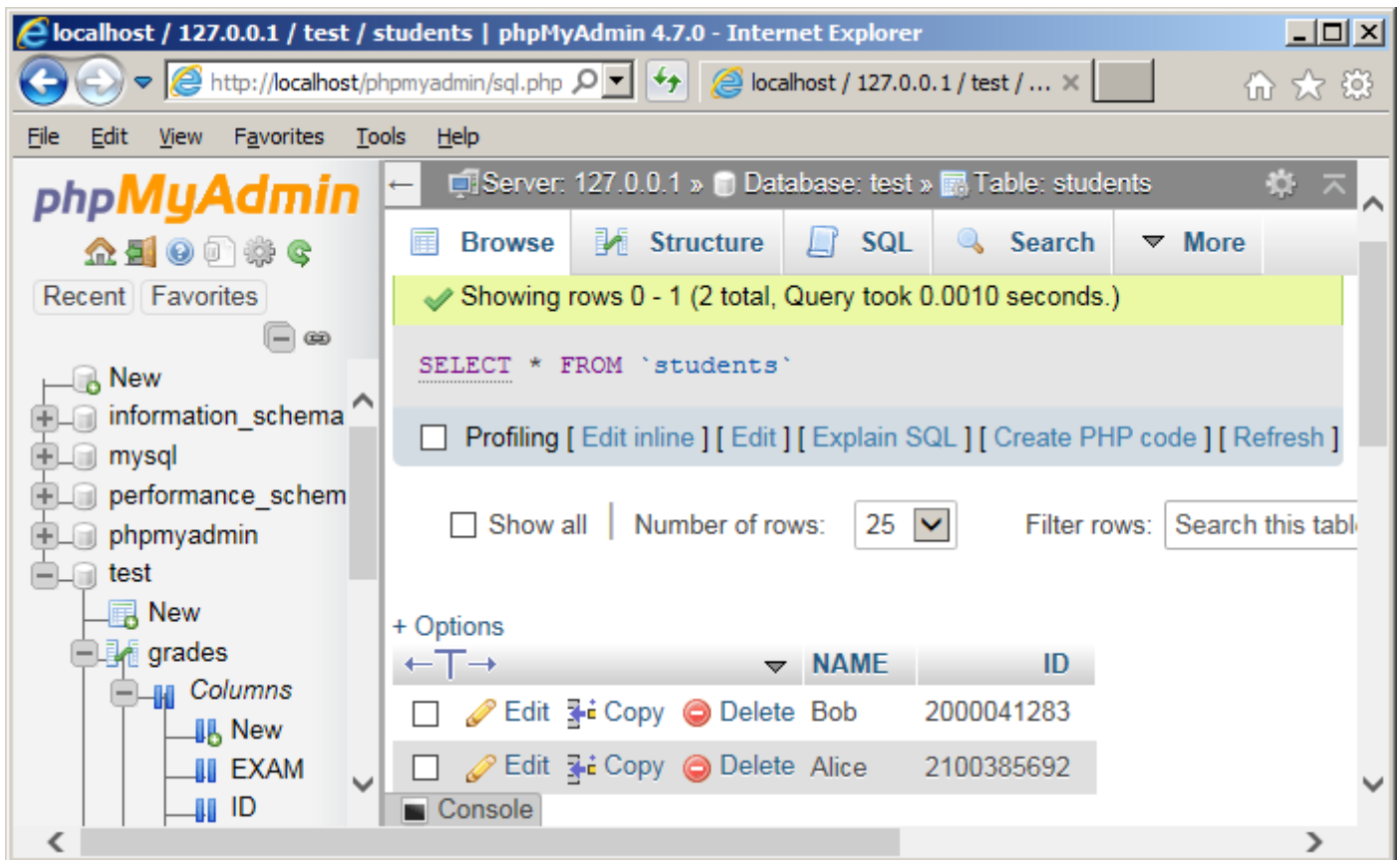
3.2 Adding Data to Tables

Click on the “Database: test” link at the top of the web page to go back to the database level, then click on the blue “SQL” link to go back to the SQL query box that you have used before.

In the SQL query box, use the SQL INSERT command to add yourself (with your name and your student ID number) and one or two of your friends to the “students” table of the “test” database. For example:

```
INSERT INTO STUDENTS VALUES ('Alice', 2100385692), ('Bob', 2000041283);
```

then click on the Go button to have the MySQL DBMS execute your code. You then get a “two rows inserted” message. Click on the blue “Structure” link to get the list of tables in the “test” database, then click on the name of the “students” table. You can then see all the data which is now in the table:



Similarly, add data to the “grades” table of the “test” database for you and your friends. Remember that each row in that table must have a student ID number, a quiz grade, a final exam grade, and a total grade. Since you do not want to have to compute the total grade yourself by hand, just use 0 for the total grade for now. For example:

```
INSERT INTO GRADES VALUES (2100385692, 89, 92, 0), (2000041283, 45, 62, 0);
```

Check the content of the “grades” table of the “test” database to make sure that the data has been correctly added to the table.

3.3 Changing Data

The total grades that you inserted into the “grades” table just above are obviously not correct. To change these total grades, we can use an SQL UPDATE command that will re-compute the total grade for a student as being the weighted average of the quiz grade (counting for 25%) and the exam grade (counting for 75%). Here is the SQL code that you can type in the SQL query box:

```
UPDATE GRADES SET TOTAL = 0.25 * QUIZ + 0.75 * EXAM;
```

Check the content of the “grades” table of the “test” database to make sure that the total grade is now correct for every student in the table. For example:

localhost / 127.0.0.1 / test / grades | phpMyAdmin 4.7.0 - Internet Explorer

http://localhost/phpmyadmin/sql.php

File Edit View Favorites Tools Help

phpMyAdmin

Recent Favorites

Server: 127.0.0.1 » Database: test » Table: grades

Browse Structure SQL Search More

Showing rows 0 - 1 (2 total, Query took 0.0010 seconds.)

`SELECT * FROM `grades``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

	ID	QUIZ	EXAM	TOTAL
☐ Edit Copy Delete	2000041283	45	62	57.75
☐ Edit Copy Delete	2100385692	89	92	91.25

Console

Exercise 1: use an SQL UPDATE command together with a WHERE clause to change your quiz grade in the “grades” table of the “test” database. The quiz grades of your friends must not change. Look at the “Databases” lecture notes to see how to do this. The result of the change should for example look like this, where the quiz grade of the student with student ID number 2000041283 has changed from 45 to 30:

localhost / 127.0.0.1 / test / grades | phpMyAdmin 4.7.0 - Internet Explorer

http://localhost/phpmyadmin/sql.php

File Edit View Favorites Tools Help

phpMyAdmin

Recent Favorites

Server: 127.0.0.1 » Database: test » Table: grades

Browse Structure SQL Search More

Showing rows 0 - 1 (2 total, Query took 0.0010 seconds.)

`SELECT * FROM `grades``

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

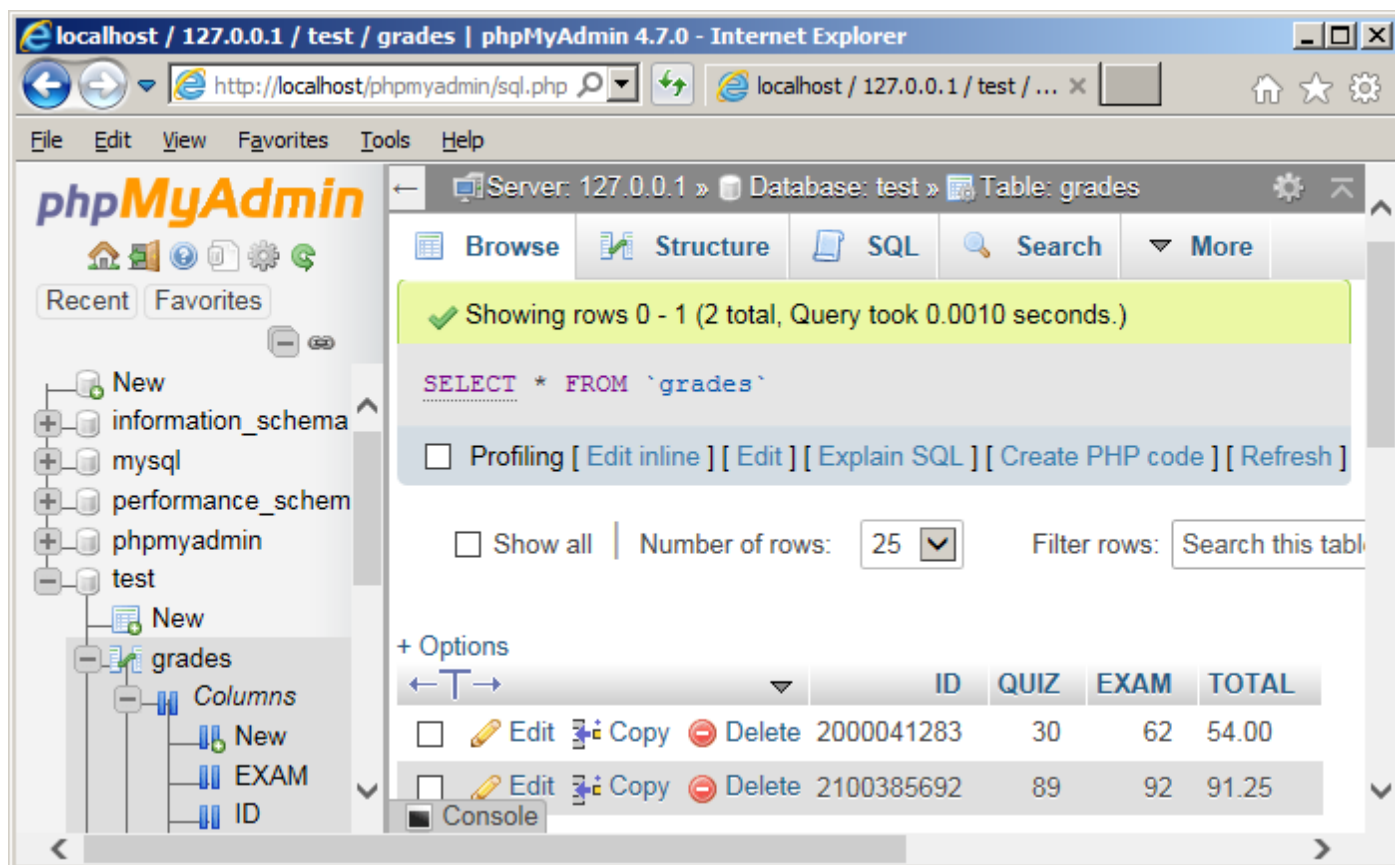
+ Options

	ID	QUIZ	EXAM	TOTAL
☐ Edit Copy Delete	2000041283	30	62	57.75
☐ Edit Copy Delete	2100385692	89	92	91.25

Console

Note how the total grade does not automatically change when you change the quiz grade: unlike what happens in a spreadsheet, formulas are not stored in the database itself and are not automatically recomputed when the data changes.

Exercise 2: use an SQL UPDATE command together with a WHERE clause to recompute your total grade in the “grades” table of the “test” database. The total grades of your friends must not change. The result of the change should for example look like this, where the total grade of the student with student ID number 2000041283 has changed from 57.75 to 54.00:

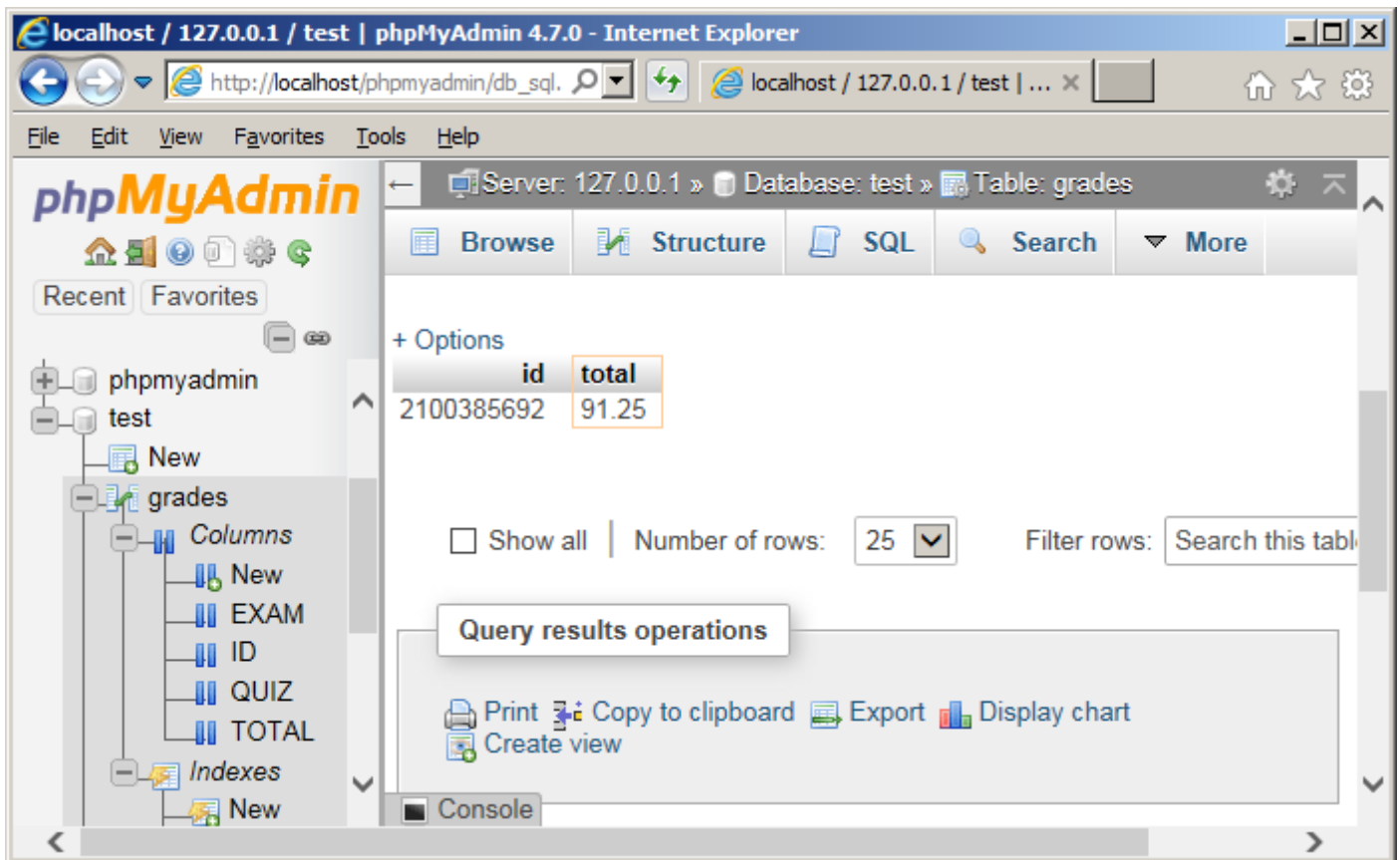


The screenshot shows the phpMyAdmin interface in Internet Explorer. The browser address bar shows 'localhost / 127.0.0.1 / test / grades | phpMyAdmin 4.7.0 - Internet Explorer'. The phpMyAdmin interface shows the 'test' database selected, and the 'grades' table is open. The table structure is visible, showing columns: ID, QUIZ, EXAM, and TOTAL. The table contains two rows of data:

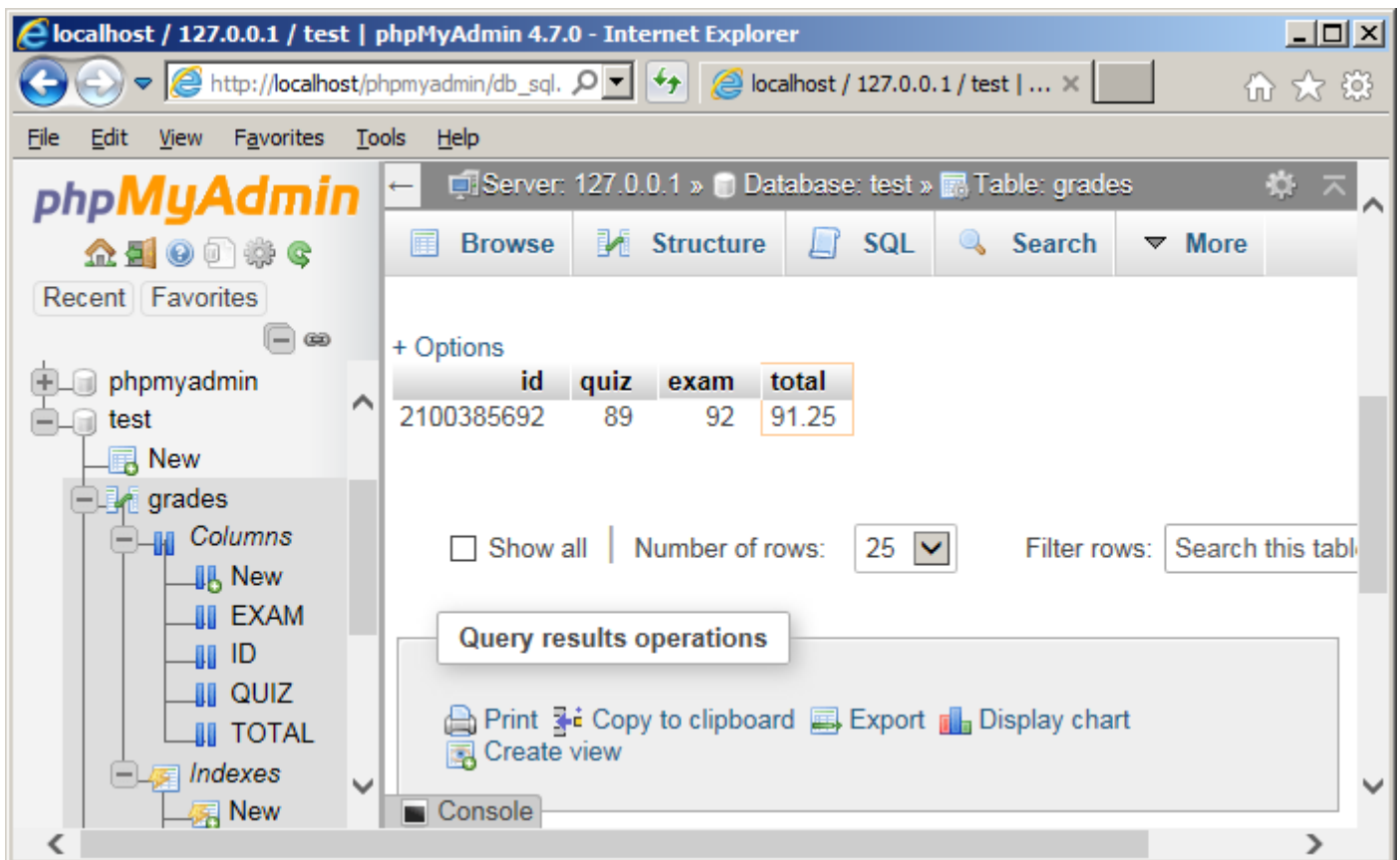
ID	QUIZ	EXAM	TOTAL
2000041283	30	62	54.00
2100385692	89	92	91.25

3.4 Searching Data

Exercise 3: use an SQL SELECT command together with a WHERE clause to get the student ID numbers and total grades of all the students with a total grade bigger than or equal to 80.0. Look at the “Databases” lecture notes to see how to do this. The result of the search should for example look like this:

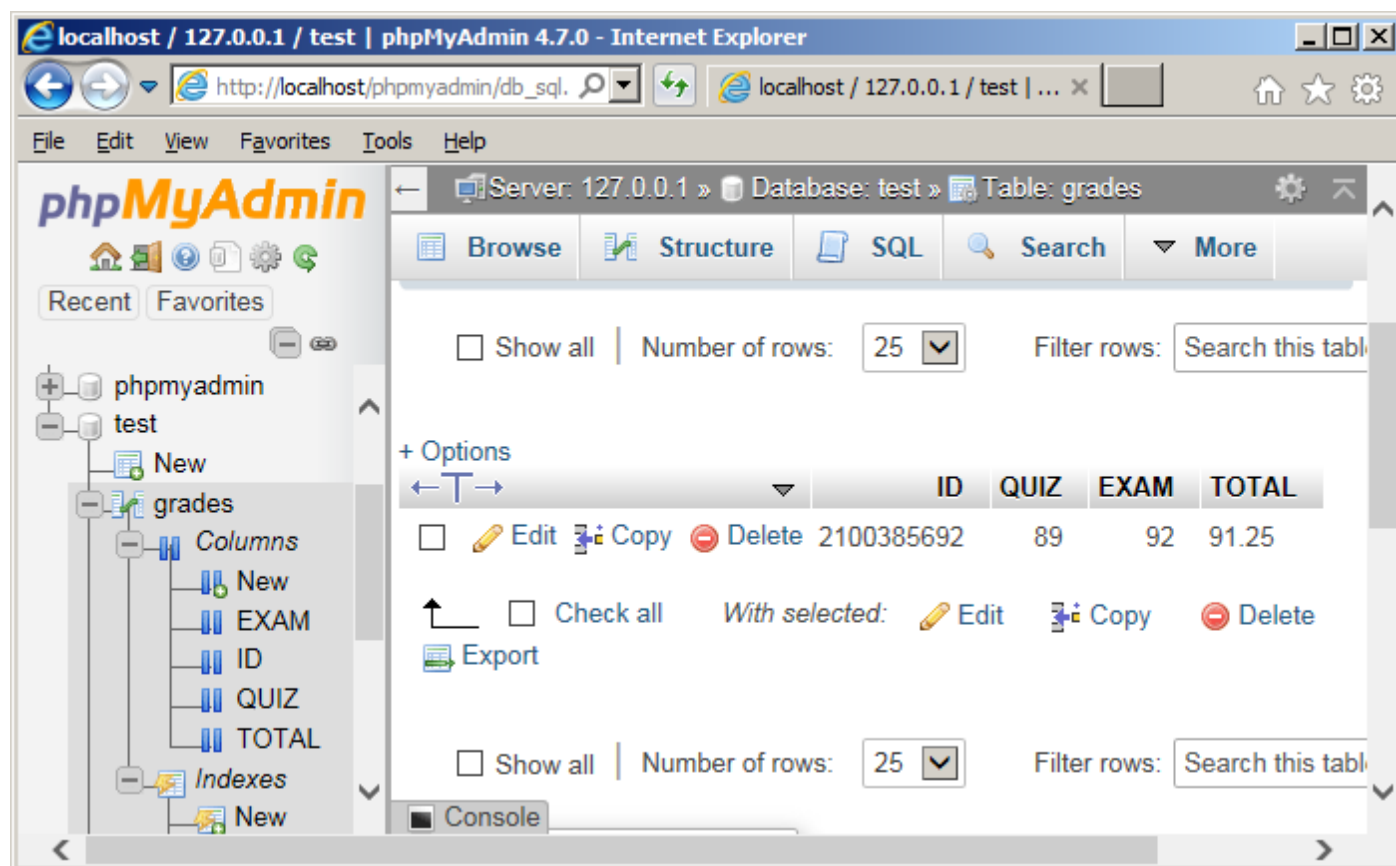


Exercise 4: redo the previous search but this time print all the information (student ID number, quiz grade, exam grade, total grade) for all the students with a total grade bigger than or equal to 80.0. The result of the search should for example look like this:



Exercise 5: redo the previous search but this time print all the information (student ID number, quiz grade, exam grade, total grade) for all the students with a total grade bigger than or equal to 80.0 by using the

star symbol * in the search query, instead of using “ID, QUIZ, EXAM, TOTAL” in your query. The start symbol * means “everything” in SQL and it is an easy way to print complete rows of data from a table without having to specify one by one the names of all the columns in the table. The result of your query should for example looks like this:

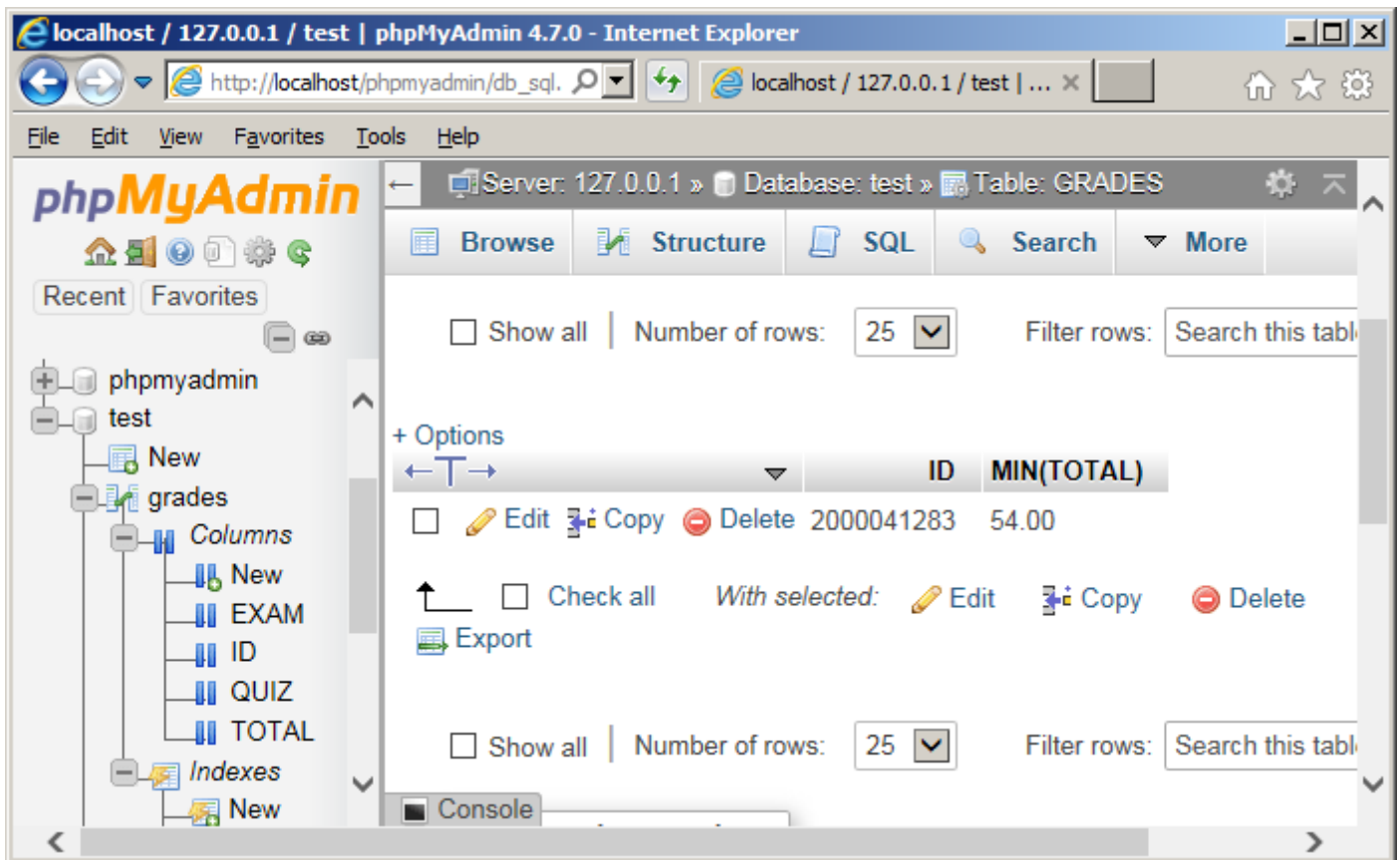


3.5 Advanced Search

The SQL programming language has many functions that you can use to do more advanced data searching and processing. For example, execute the following SQL code in the SQL query box to compute the student ID number and total grade of the student with the lowest total grade:

```
SELECT ID, MIN(TOTAL) FROM GRADES;
```

The result should then look like this:

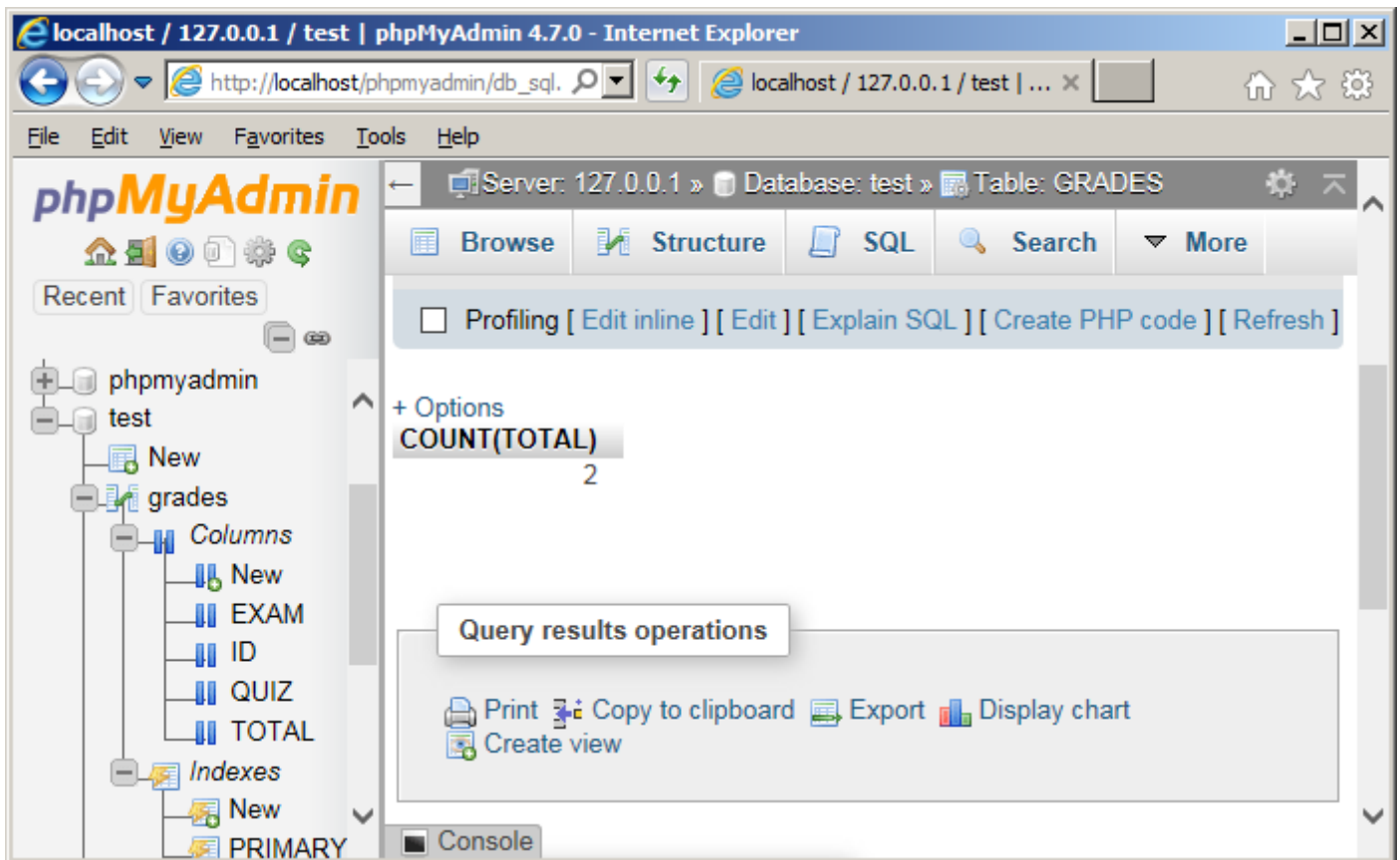


Exercise 6: similarly, use the SQL MAX function to compute the student ID number and total grade of the student with the highest total grade.

Execute the following SQL code that uses the COUNT function to count how many students have a total grade between 50 and 100:

```
SELECT COUNT(TOTAL) FROM GRADES WHERE TOTAL BETWEEN 50.0 AND 100.0;
```

The result should then look like this:

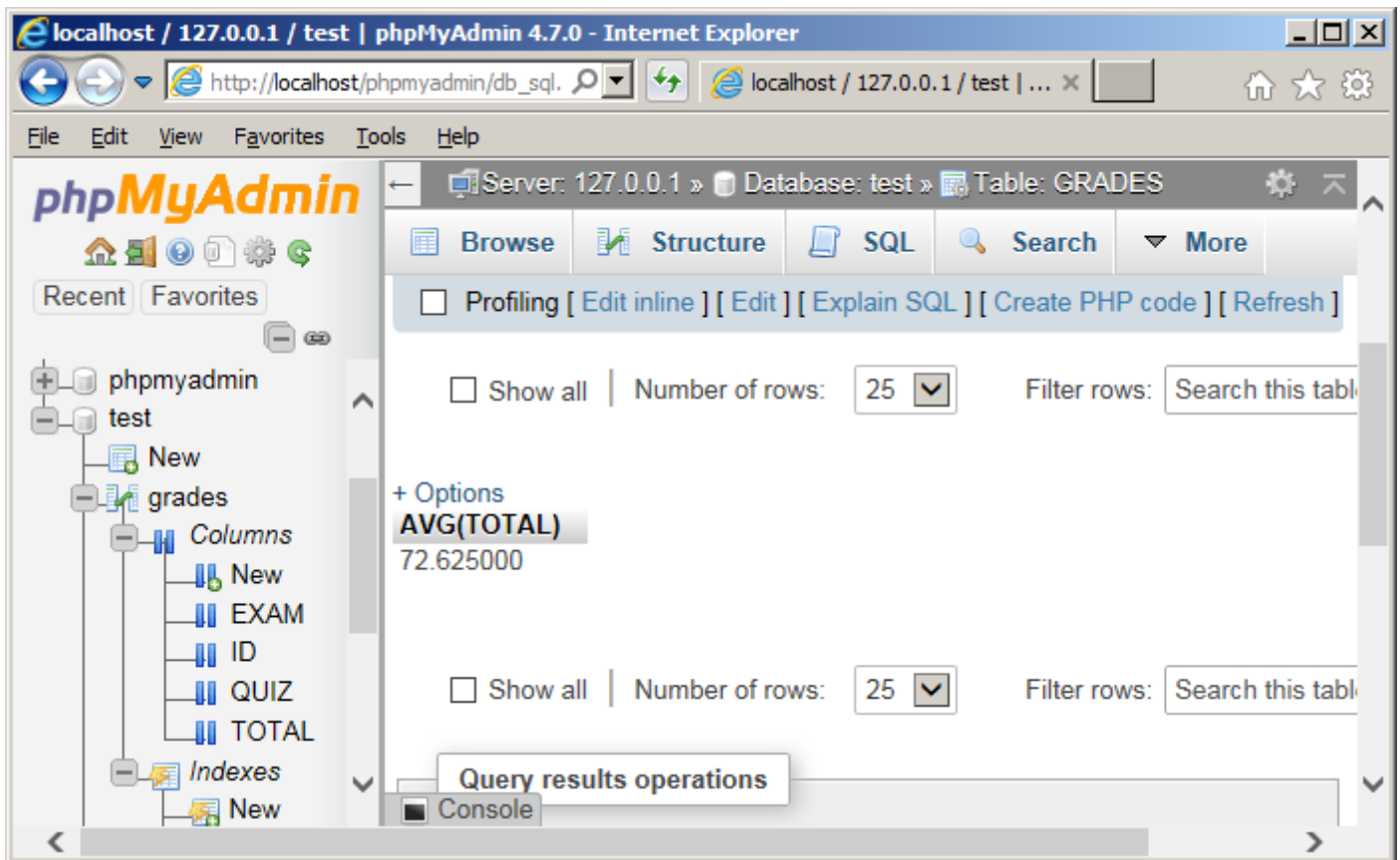


Exercise 7: similarly, use the COUNT function and the WHERE ... BETWEEN operator to count how many students have a low quiz grade (a quiz grade below 50.0).

Execute the following SQL code to compute the average total grade for all the students:

```
SELECT AVG(TOTAL) FROM GRADES;
```

The result should then look like this:

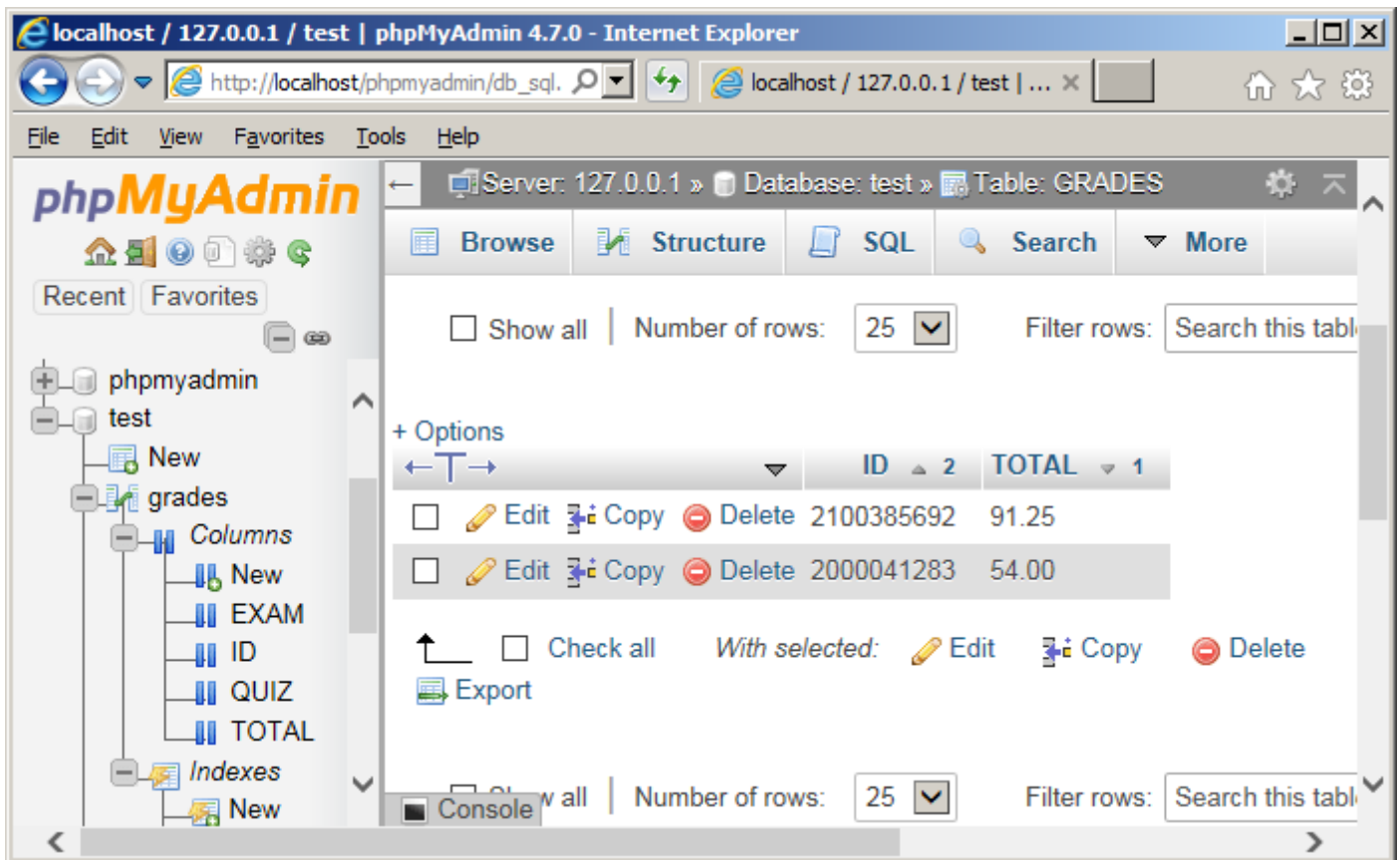


Exercise 8: similarly, use the AVG function multiple times inside a single SQL command to compute at the same time the average quiz grade, the average exam grade, and the average total grade for all the students.

Execute the following SQL code that uses the SORT function to print the student ID number and total grades for all the students, sorted first by descending (decreasing) total grade, then sorted by ascending (increasing) student ID number (if two students have the same total grade):

```
SELECT ID, TOTAL FROM GRADES ORDER BY TOTAL DESC, ID ASC;
```

The result should then look like this:



We now want to sort again the same data, in the same order, but we also want to see in the output the names of the students, not just the student ID numbers. This is more complicated because we are sorting the total grades from the “grades” table but the names of the students are in the separate “students” table. So if we want to see in the output both the sorted grades and the names of the students then we need to combine information from two separate tables. This is called doing a “join”: merging information from two separate tables to compute a single result.

Here is the code to do the SQL join between the two tables:

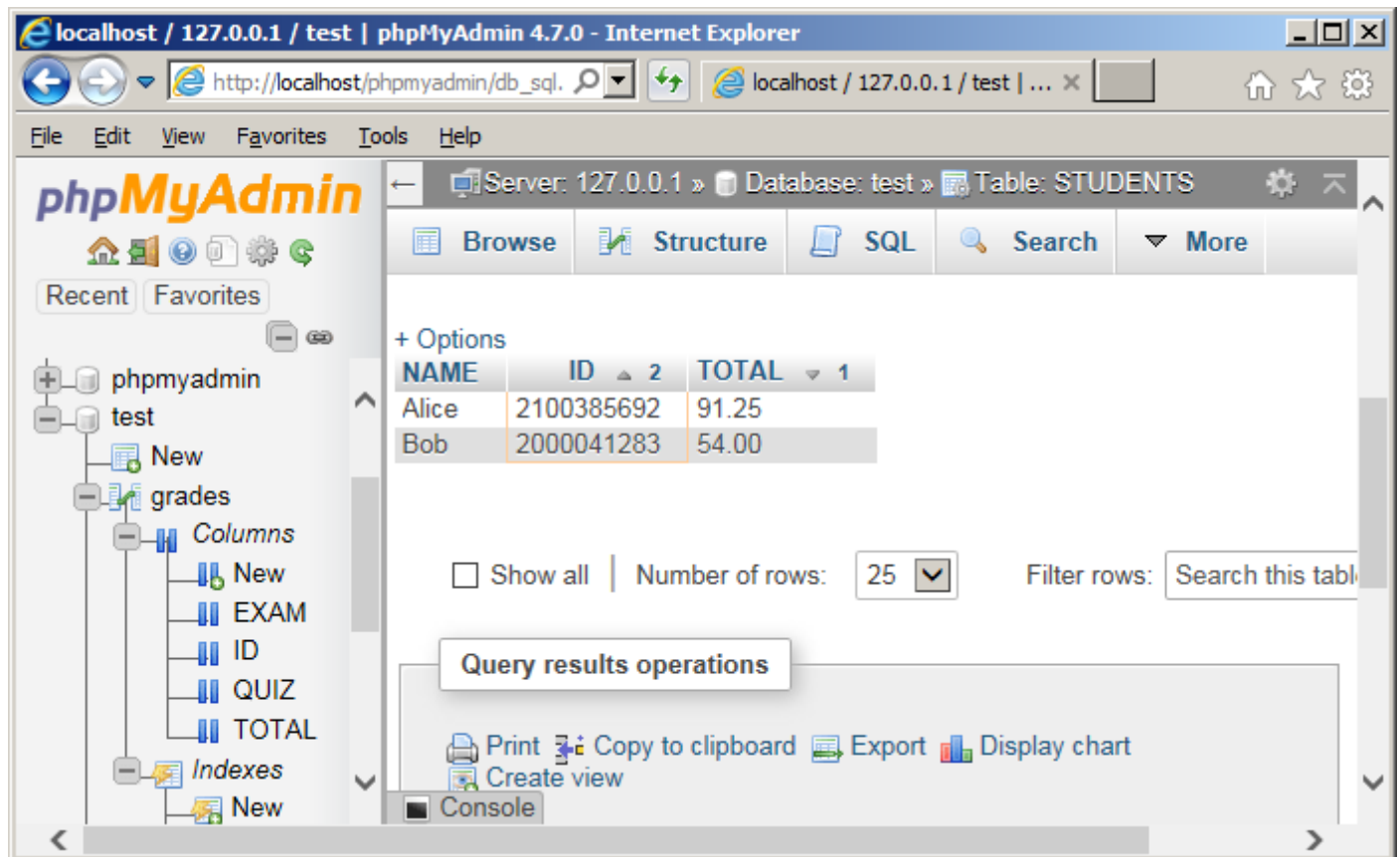
```
SELECT NAME, STUDENTS.ID, TOTAL FROM STUDENTS, GRADES
WHERE STUDENTS.ID = GRADES.ID
ORDER BY TOTAL DESC, ID ASC;
```

In this SQL code, we indicate that we want to print the student names, student ID numbers, and total grades, and that this information comes from two separate tables: the “students” table and the “grades” table. Since the “students” table and the “grades” table both have a column named “ID”, we need to specify that we want to print the data coming from the “ID” column of the “students” table, not from the “ID” column of the “grades” table, which is why we specify “STUDENTS.ID” to print the student ID numbers, not just “ID” (even though both columns contain the exact same data, SQL will be confused if we do not specify exactly which column of data we want to print).

Since the information we want to print comes from two separate tables, we have to tell SQL how to match the data coming from one table with the data coming from the other table. We do this by using a WHERE clause to tell SQL that data coming from the “students” table matches data coming from the “grades” table if and only if the data from the “students” table has a student ID number which is equal to the student ID number for the data coming from the “grades” table. This is the part of the code that correctly does the join between the different pieces of data coming from the two different tables.

In the last part of the code, we simply sort the result of the join in the same way we sorted total grades in the previous search (see above).

Execute the code given above for the SQL join. The result should then look like this:

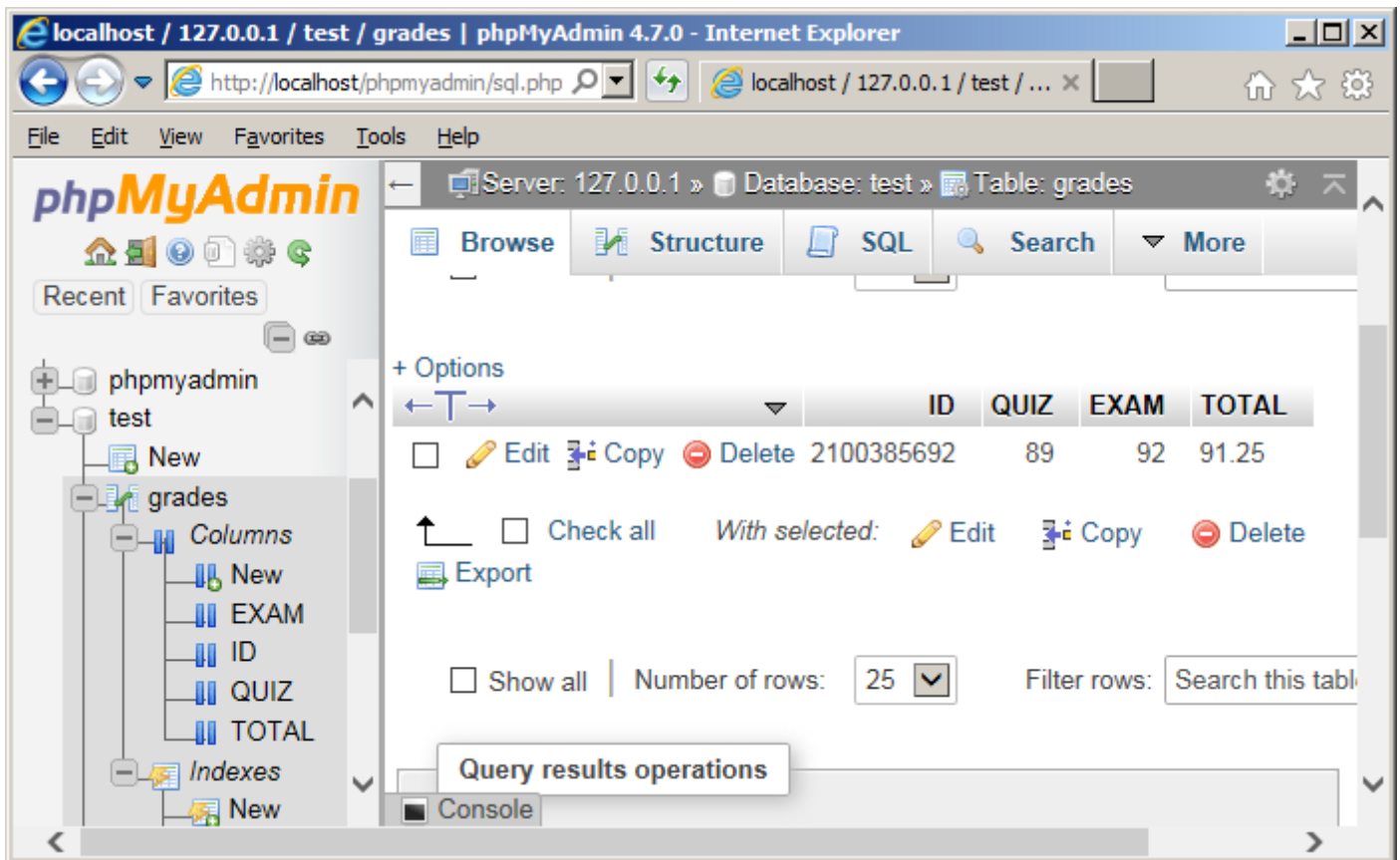


3.6 Deleting Data from Tables

To delete one or more rows of data from a table, use the SQL DELETE command together with a WHERE clause specifying which rows to delete. For example, to delete from the “grades” table the data for the student with student ID number 2000041283, you can use the following code:

```
DELETE FROM GRADES WHERE ID = 2000041283;
```

Modify and execute this code to delete your own grade information from your “grades” table. Click on the “Database: test” link at the top of the web page, then click on the name of the “grades” table to verify that your own grade data has been deleted from your “grades” table. For example:



Note that this does change the content of the “students” table! Click on the “Database: test” link at the top of the web page, then click on the name of the “students” table and you will see that your name and student ID number are still there!

Exercise 9: write and execute some SQL code to delete your name and student ID number from the “students” table. Check that the deletion actually happened.

3.7 Deleting Tables

To delete a complete table at once, simply use the SQL DROP command. For example, to delete the “grades” table, execute the following code:

```
DROP TABLE GRADES;
```

Then click on the blue “Structure” link in the web page to verify that the “grades” table has been deleted.

Exercise 10: similarly, delete the “students” table. Verify that the “test” database is then completely empty.

4 Shutting Down

In the first section, you started XAMPP, then used XAMPP to start MySQL and Apache, then started a web browser. When shutting the whole system down, simply stop all the different pieces of software in the reverse order: close the web browser, use the XAMPP Control Panel to stop Apache, use the XAMPP Control Panel to stop MySQL, then close the XAMPP Control Panel by using the Quit button of the XAMPP Control Panel (do not use the little Microsoft Windows “X” button in the upper right corner of the XAMPP Control Panel window: this “X” button does not stop XAMPP, it only hides it and XAMPP will keep running!)

Once you have shut down everything, remember to upload your answers to the 10 exercises on iSpace.